

Inferential Feature Analysis: Latino Trump Support (CMPS 2024)

Analysis Pipeline — Random Forest + SHAP + Bootstrap

J. Grijalva

2026-04-22

Table of contents

1	Analysis Parameters	2
2	Data and Sample Construction	2
3	Preprocessing	2
4	Train/Test Split and Imputation	2
5	Tier Construction and Partisan Leakage Audit	2
6	Model Fitting	3
7	Benchmark Models	3
8	SHAP Computation	4
9	Feature Correlation Structure	4
10	Bootstrap Confidence Intervals	4
11	Results Assembly	5
12	Threshold Sensitivity	5
13	Repeated-Resampling Robustness	5
14	Main Results	6
14.1	Model Performance	6
14.2	Stable Non-Partisan Features	6
14.3	Figure 1. Feature Importance (Beeswarm)	7
14.4	Figure 2. Bootstrap Confidence Intervals	8
14.5	Figure 3. SHAP Dependence (Top Feature)	9

Table 1: Analysis Configuration.

Parameter	Value
Bootstrap iterations (B)	500
Trees per forest	500
Test proportion	0.25
Rank stability threshold (k)	3
Development mode	FALSE
Raw data file	cmps2024_fulladult_112425 .csv

Table 2: Latino Subsample Summary.

Item	Value
Latino N	3051
Trump voters	1286
Trump %	42.2%
Weight range	[0.08, 25.99]

15 Save Results Bundle **9**

16 Session Information **9**

1 Analysis Parameters

2 Data and Sample Construction

3 Preprocessing

4 Train/Test Split and Imputation

5 Tier Construction and Partisan Leakage Audit

Exclusion-list correction (2026-04-21, first pass). The Tier 2 and Tier 3 exclusion sets were revised to correct three implementation issues identified during a post-hoc audit against the pre-registered exclusion criteria. First, the tautological set listed **q3q** for the Jan 6 insurrectionists feeling thermometer, but the CSV column is **q3r17** — the entry matched no column and the thermometer was silently retained. Second, **q175** (approve of Trump defying court orders) directly evaluates Trump’s own behavior and falls under the pre-registered tautology criterion for items

Table 3: Train/Test Split Summary.

Set	N	Trump %	Features
Training	2289	41.5	502
Test	762	44.0	502

Table 4: Progressive Exclusion Tiers.

Tier	Train N	Test N	Features	Excluded
1: Full	2289	762	502	0
2: Non-Tautological	2289	762	476	26
3: Non-Partisan (Ring 1)	2289	762	460	42

Table 5: Tier 3 AUC Across Learners.

Learner	Test AUC
Random Forest	0.919
Elastic Net Logistic	0.915
XGBoost	0.932

that evaluate the Trump presidency; it was added to the tautological set. Third, the partisan set now includes q28r1–q28r5 (party and civic GOTV contact flags), mirroring the 2020 analysis’s Q28R1–Q28R19 exclusion; these items are partisan-by-construction and their omission had made q28r2 (GOP GOTV contact) the top Tier 3 feature in the uncorrected run.

Exclusion-list correction (2026-04-21, second pass — top-20 audit). A subsequent audit of the corrected top-20 Tier 3 features against the instrument identified three additional items that satisfied the pre-registered exclusion criteria but had been retained. Under the tautology criterion (“response options directly evaluate Trump’s personal performance, responsibility, or trustworthiness”), q2 (“When it comes to democracy today, do you think the President and the executive branch are: Using too much control / the right amount of control / Not using enough control”) was moved to the tautological set because CMPS 2024 was fielded after the Trump inauguration, making “the President” refer to Trump and making the response options direct evaluations of his executive-power use; this item is structurally parallel to the already-excluded q175. Also under the tautology criterion, q174 (“approve or disapprove of the U.S. President using the 1798 Alien Enemies Act to arrest, detain, and deport immigrants...”) was moved to the tautological set as a direct approval item for a Trump-specific action, again structurally parallel to q175. Under the partisan criterion, q45 (“How hard do you think the Republican Party really works on issues Black people care about?”) was moved to the partisan exclusion set as a direct performance evaluation of the Republican Party as a partisan actor, parallel in construction to the already-excluded q3r1–q3r3 thermometers toward Democrats, Republicans, and Independents. The corrected results below are the reported ones; earlier uncorrected runs are retained as backup artifacts for transparency.

6 Model Fitting

7 Benchmark Models

Elastic net logistic regression and XGBoost are fitted on the same Tier 3 train/test split to confirm that findings are a property of the data rather than an artifact of Random Forest.

8 SHAP Computation

9 Feature Correlation Structure

When features are correlated, SHAP distributes predictive credit across correlated items in ways that are mathematically principled but may not map onto standalone constructs. The correlation matrix below documents the pairwise relationships among the top features.

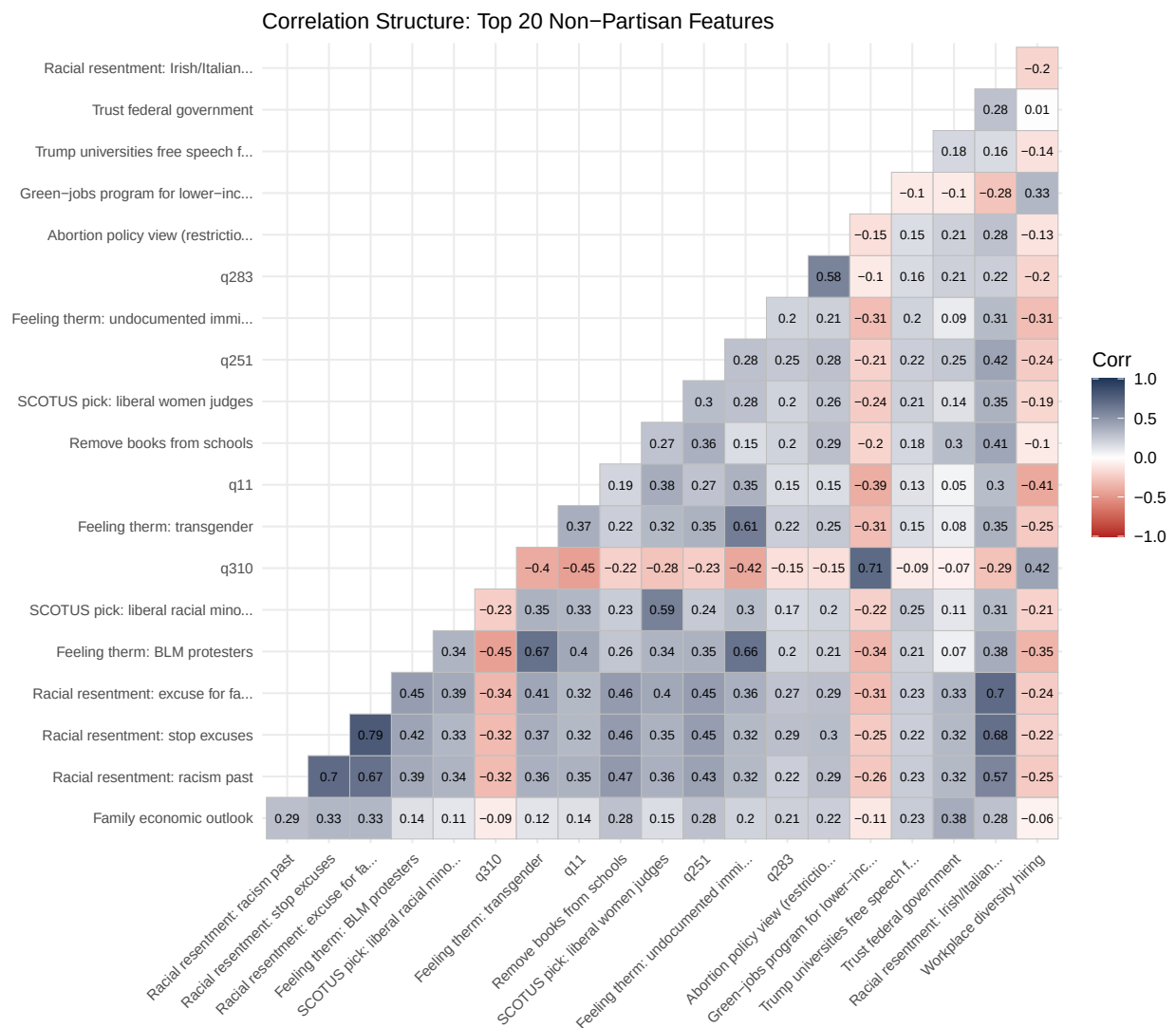


Figure 1: Pairwise Pearson correlations among top 20 Tier 3 features (test set).

10 Bootstrap Confidence Intervals

The bootstrap resamples the training set 500 times, refitting a Random Forest and computing TreeSHAP on the fixed test set each time. This isolates model instability from evaluation-set

Table 6: Threshold Sensitivity: Stable Features at $k = 2.0, 3.0$, and 4.0 .

Rank	Feature	sigma_j	k=2.0	k=3.0	k=4.0
1	Family economic outlook	0.99	\checkmark	\checkmark	\checkmark
2	Racial resentment: racism past	1.02	\checkmark	\checkmark	\checkmark
3	Racial resentment: stop excuses	1.01	\checkmark	\checkmark	\checkmark
4	Racial resentment: excuse for failure	0.99	\checkmark	\checkmark	\checkmark
5	Feeling therm: BLM protesters	1.28	\checkmark	\checkmark	\checkmark
6	SCOTUS pick: liberal racial minority judges	3.35			\checkmark
7	q310	2.85		\checkmark	\checkmark
8	Feeling therm: transgender	3.76			\checkmark
9	q11	5.00			
10	Remove books from schools	4.14			
11	SCOTUS pick: liberal women judges	6.28			
12	q251	4.74			
13	Feeling therm: undocumented immigrants	4.76			
14	q283	6.76			
15	Abortion policy view (restriction level)	5.50			
16	Green-jobs program for lower-income workers	6.27			
17	Trump universities free speech framing	6.88			
18	Trust federal government	10.28			
19	Racial resentment: Irish/Italian comparison	6.79			
20	Workplace diversity hiring	7.71			

composition variance. Each feature’s 95% CI is the 2.5th–97.5th percentile of the bootstrap distribution of $E_w[|\phi_j|]$.

11 Results Assembly

12 Threshold Sensitivity

The stability threshold k is a pre-specified reporting rule, not a universal standard. The table below shows which features qualify as stable findings at $k = 2.0$, $k = 3.0$, and $k = 4.0$, holding the magnitude criterion ($ci_lo > 0$) constant throughout.

13 Repeated-Resampling Robustness

The bootstrap assesses training-sample sensitivity conditional on one fixed test partition. To confirm the feature architecture is not an artifact of the particular split, five independent 75/25 partitions are drawn and the full Tier 3 RF + SHAP pipeline is run on each. Spearman rank correlations of feature importance across all pairs test whether the same features surface regardless of the split.

Table 7: Spearman Rank Correlations Across 5 Independent Splits (min = `r sprintf('%.3f', min_rho)`, mean = `r sprintf('%.3f', mean_rho)`).

Split	V1	V2	V3	V4	V5
1	1.000	0.940	0.938	0.950	0.933
2	0.940	1.000	0.942	0.943	0.944
3	0.938	0.942	1.000	0.941	0.940
4	0.950	0.943	0.941	1.000	0.934
5	0.933	0.944	0.940	0.934	1.000

Table 8: AUC Across the Progressive Exclusion Framework.

Model	Test AUC	Drop from T1	Boot Mean AUC	95% CI
Tier 1 (Full)	0.990	NA	NA	—
Tier 2 (Non-Tautological)	0.972	0.018	NA	—
Tier 3 (Non-Partisan)	0.919	0.070	0.912	[0.904, 0.92]

14 Main Results

14.1 Model Performance

14.2 Stable Non-Partisan Features

Features are reported as stable findings under a conjunctive decision rule pre-specified before examining results: the bootstrap 95% CI lower bound for $E_w[|\phi_j|]$ must exceed zero (magnitude criterion) **and** $\sigma_j \leq 3$ (rank stability criterion). The first condition ensures the feature’s importance is bounded away from nothing across 500 training resamples. The second ensures its ordinal position in the importance ranking is stable enough to support comparative claims.

Table 9: Stable Ring 1 Non-Partisan Features (`ci_lo > 0`, `sigma_j <= 3`).

Rank	Feature	Code	$E_w[\phi]$	CI lo	CI hi	σ_j
1	Family economic outlook	q30	0.03322	0.02159	0.03556	0.99
2	Racial resentment: racism past	q176	0.02604	0.01955	0.03194	1.02
3	Racial resentment: stop excuses	q177	0.02484	0.01750	0.03038	1.01
4	Racial resentment: excuse for failure	q180	0.02388	0.01928	0.03019	0.99
5	Feeling therm: BLM protesters	q3r18	0.01772	0.01126	0.02108	1.28
7	q310	q310	0.01280	0.00798	0.01780	2.85

14.3 Figure 1. Feature Importance (Beeswarm)

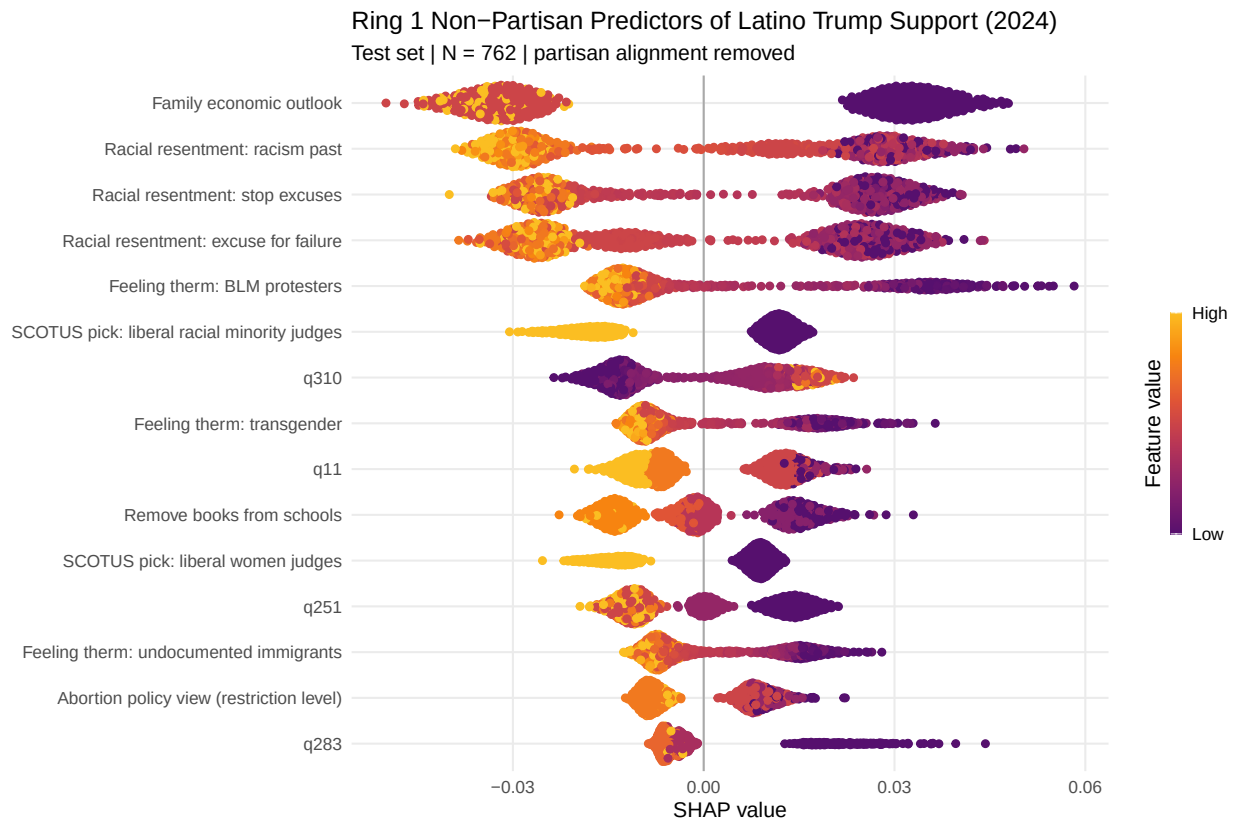


Figure 2

14.4 Figure 2. Bootstrap Confidence Intervals

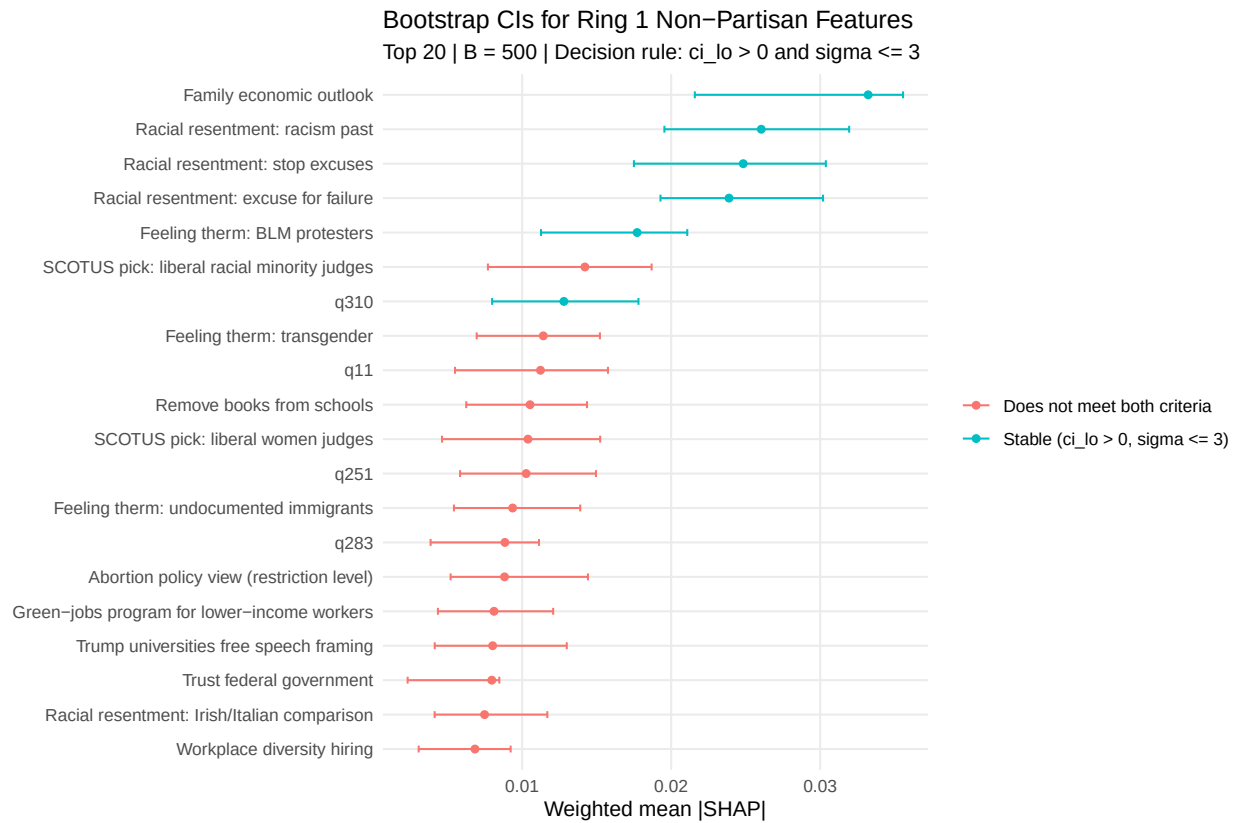


Figure 3

14.5 Figure 3. SHAP Dependence (Top Feature)

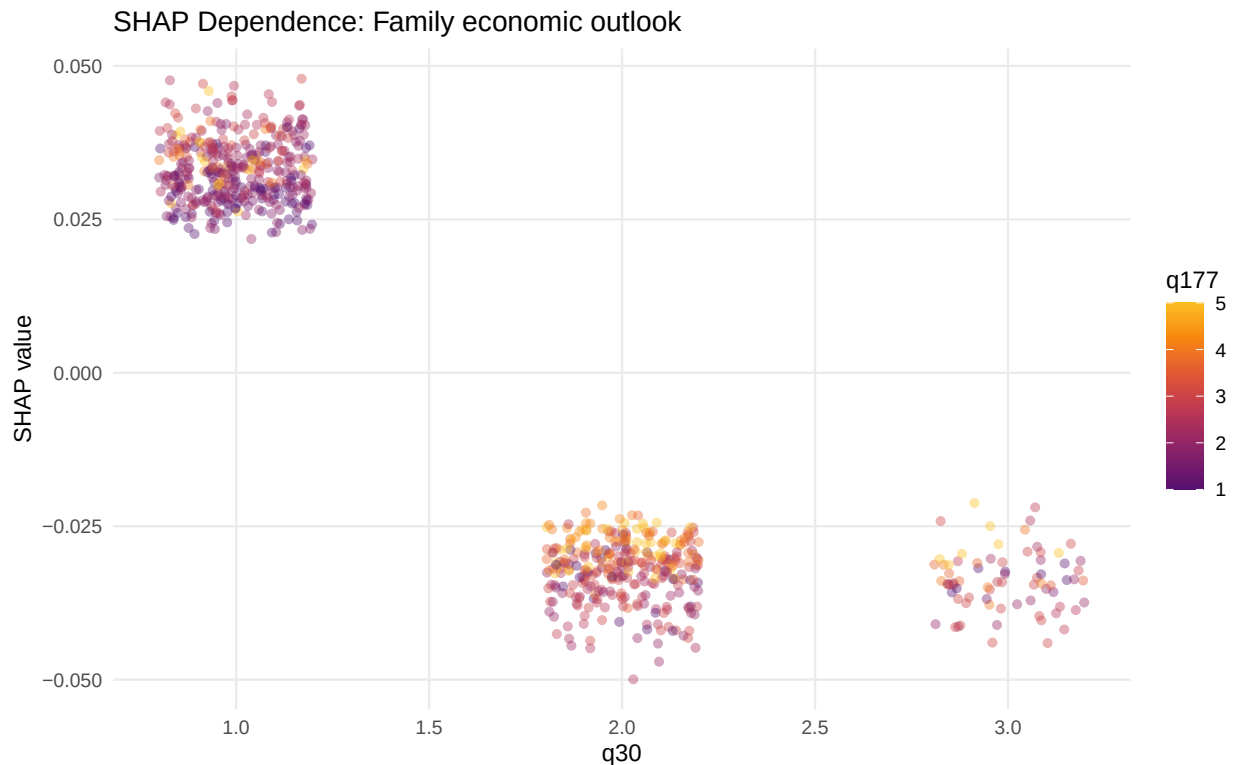


Figure 4

15 Save Results Bundle

All model objects, SHAP matrices, bootstrap results, tier metadata, and summary tables are saved to a single `.rds` file. The reporting QMD loads this file to generate all manuscript figures and tables without re-running the model.

The results bundle has been saved to `ifa_results.rds` (17.1 MB). Load this file in the reporting QMD with `readRDS("ifa_results.rds")`.

16 Session Information

```
- Session info -----
setting  value
version  R version 4.5.3 (2026-03-11)
os       macOS Tahoe 26.4.1
system   aarch64, darwin20
ui       X11
language (EN)
collate  en_US.UTF-8
```

```

ctype    en_US.UTF-8
tz        America/Indiana/Indianapolis
date      2026-04-22
pandoc    3.6.3 @ /Applications/RStudio.app/Contents/Resources/app/quarto/bin/tools/aarch64/ (
quarto    1.8.25 @ /Applications/RStudio.app/Contents/Resources/app/quarto/bin/quarto

```

```

- Packages -----
package      * version      date (UTC) lib source
bit           4.6.0         2025-03-06 [1] CRAN (R 4.5.0)
bit64         4.6.0-1       2025-01-16 [1] CRAN (R 4.5.0)
caret        * 7.0-1         2024-12-10 [1] CRAN (R 4.5.0)
class        7.3-23        2025-01-01 [2] CRAN (R 4.5.3)
cli           3.6.5         2025-04-23 [1] CRAN (R 4.5.0)
codetools    0.2-20        2024-03-31 [2] CRAN (R 4.5.3)
crayon        1.5.3         2024-06-20 [1] CRAN (R 4.5.0)
data.table   1.18.2.1      2026-01-27 [1] CRAN (R 4.5.2)
digest       * 0.6.39       2025-11-19 [1] CRAN (R 4.5.2)
dplyr        * 1.2.0         2026-02-03 [1] CRAN (R 4.5.2)
evaluate      1.0.5         2025-08-27 [1] CRAN (R 4.5.0)
farver        2.1.2         2024-05-13 [1] CRAN (R 4.5.0)
fastDummies  * 1.7.5         2025-01-20 [1] CRAN (R 4.5.0)
fastmap       1.2.0         2024-05-15 [1] CRAN (R 4.5.0)
forcats      * 1.0.1         2025-09-25 [1] CRAN (R 4.5.0)
foreach       1.5.2         2022-02-02 [1] CRAN (R 4.5.0)
future        1.69.0        2026-01-16 [1] CRAN (R 4.5.2)
future.apply  1.20.2        2026-02-20 [1] CRAN (R 4.5.2)
generics      0.1.4         2025-05-09 [1] CRAN (R 4.5.0)
ggcorrplot   * 0.1.4.1      2023-09-05 [1] CRAN (R 4.5.0)
ggplot2      * 4.0.2         2026-02-03 [1] CRAN (R 4.5.2)
glmnet       * 4.1-10        2025-07-17 [1] CRAN (R 4.5.0)
globals      0.19.1        2026-03-13 [1] CRAN (R 4.5.2)
glue          1.8.0         2024-09-30 [1] CRAN (R 4.5.0)
gower        1.0.2         2024-12-17 [1] CRAN (R 4.5.0)
gtable       0.3.6         2024-10-25 [1] CRAN (R 4.5.0)
hardhat      1.4.2         2025-08-20 [1] CRAN (R 4.5.0)
hms          1.1.4         2025-10-17 [1] CRAN (R 4.5.0)
htmltools    0.5.9         2025-12-04 [1] CRAN (R 4.5.2)
ipred        0.9-15        2024-07-18 [1] CRAN (R 4.5.0)
iterators    1.0.14        2022-02-05 [1] CRAN (R 4.5.0)
jsonlite     2.0.0         2025-03-27 [1] CRAN (R 4.5.0)
kableExtra   * 1.4.0         2024-01-24 [1] CRAN (R 4.5.0)
knitr        * 1.51          2025-12-20 [1] CRAN (R 4.5.2)
labeling     0.4.3         2023-08-29 [1] CRAN (R 4.5.0)
lattice      * 0.22-9        2026-02-09 [2] CRAN (R 4.5.3)
lava         1.8.2         2025-10-30 [1] CRAN (R 4.5.0)
lifecycle    1.0.5         2026-01-08 [1] CRAN (R 4.5.2)
listenv      0.10.1        2026-03-10 [1] CRAN (R 4.5.2)
lubridate    * 1.9.5         2026-02-04 [1] CRAN (R 4.5.2)

```

magrittr	2.0.4	2025-09-12	[1]	CRAN	(R 4.5.0)
MASS	7.3-65	2025-02-28	[2]	CRAN	(R 4.5.3)
Matrix	* 1.7-4	2025-08-28	[2]	CRAN	(R 4.5.3)
missRanger	* 2.6.1	2024-12-07	[1]	CRAN	(R 4.5.0)
ModelMetrics	1.2.2.2	2020-03-17	[1]	CRAN	(R 4.5.0)
nlme	3.1-168	2025-03-31	[2]	CRAN	(R 4.5.3)
nnet	7.3-20	2025-01-01	[2]	CRAN	(R 4.5.3)
otel	0.2.0	2025-08-29	[1]	CRAN	(R 4.5.0)
parallelly	1.46.1	2026-01-08	[1]	CRAN	(R 4.5.2)
pillar	1.11.1	2025-09-17	[1]	CRAN	(R 4.5.0)
pkgconfig	2.0.3	2019-09-22	[1]	CRAN	(R 4.5.0)
plyr	1.8.9	2023-10-02	[1]	CRAN	(R 4.5.0)
pROC	* 1.19.0.1	2025-07-31	[1]	CRAN	(R 4.5.0)
prodlim	2026.03.11	2026-03-11	[1]	CRAN	(R 4.5.2)
purrr	* 1.2.1	2026-01-09	[1]	CRAN	(R 4.5.2)
R6	2.6.1	2025-02-15	[1]	CRAN	(R 4.5.0)
ranger	* 0.18.0	2026-01-16	[1]	CRAN	(R 4.5.2)
RColorBrewer	1.1-3	2022-04-03	[1]	CRAN	(R 4.5.0)
Rcpp	1.1.1	2026-01-10	[1]	CRAN	(R 4.5.2)
readr	* 2.2.0	2026-02-19	[1]	CRAN	(R 4.5.2)
recipes	1.3.1	2025-05-21	[1]	CRAN	(R 4.5.0)
reshape2	1.4.5	2025-11-12	[1]	CRAN	(R 4.5.0)
rlang	1.1.7	2026-01-09	[1]	CRAN	(R 4.5.2)
rmarkdown	2.30	2025-09-28	[1]	CRAN	(R 4.5.0)
rpart	4.1.24	2025-01-07	[2]	CRAN	(R 4.5.3)
rstudioapi	0.18.0	2026-01-16	[1]	CRAN	(R 4.5.2)
S7	0.2.1	2025-11-14	[1]	CRAN	(R 4.5.2)
scales	1.4.0	2025-04-24	[1]	CRAN	(R 4.5.0)
sessioninfo	* 1.2.3	2025-02-05	[1]	CRAN	(R 4.5.0)
shape	1.4.6.1	2024-02-23	[1]	CRAN	(R 4.5.0)
shapviz	* 0.10.3	2025-10-13	[1]	CRAN	(R 4.5.0)
stringi	1.8.7	2025-03-27	[1]	CRAN	(R 4.5.0)
stringr	* 1.6.0	2025-11-04	[1]	CRAN	(R 4.5.0)
survival	3.8-6	2026-01-16	[2]	CRAN	(R 4.5.3)
svglite	2.2.2	2025-10-21	[1]	CRAN	(R 4.5.0)
systemfonts	1.3.2	2026-03-05	[1]	CRAN	(R 4.5.2)
textshaping	1.0.5	2026-03-06	[1]	CRAN	(R 4.5.2)
tibble	* 3.3.1	2026-01-11	[1]	CRAN	(R 4.5.2)
tidyr	* 1.3.2	2025-12-19	[1]	CRAN	(R 4.5.2)
tidyselect	1.2.1	2024-03-11	[1]	CRAN	(R 4.5.0)
tidyverse	* 2.0.0	2023-02-22	[1]	CRAN	(R 4.5.0)
timechange	0.4.0	2026-01-29	[1]	CRAN	(R 4.5.2)
timeDate	4052.112	2026-01-28	[1]	CRAN	(R 4.5.2)
treeshap	* 0.4.0	2026-03-19	[1]	Github	(ModelOriented/treeshap@e287dc3)
tzdb	0.5.0	2025-03-15	[1]	CRAN	(R 4.5.0)
vctrs	0.7.1	2026-01-23	[1]	CRAN	(R 4.5.2)
viridisLite	0.4.3	2026-02-04	[1]	CRAN	(R 4.5.2)
vroom	1.7.0	2026-01-27	[1]	CRAN	(R 4.5.2)

withr	3.0.2	2024-10-28	[1]	CRAN	(R 4.5.0)
xfun	0.56	2026-01-18	[1]	CRAN	(R 4.5.2)
xgboost	* 3.2.0.1	2026-02-10	[1]	CRAN	(R 4.5.2)
xml2	1.5.2	2026-01-17	[1]	CRAN	(R 4.5.2)
yaml	2.3.12	2025-12-10	[1]	CRAN	(R 4.5.2)

[1] /Users/jessala/Library/R/arm64/4.5/library

[2] /Library/Frameworks/R.framework/Versions/4.5-arm64/Resources/library

* -- Packages attached to the search path.
